

COURSE 3 - DATA ENGINEERING 2026**C3-074A - MeterStream Streaming Gate**

Protected independent phase Gate A.

Artifact	C3-074A
Status	2026 - LEARNER EDITION
Phase	C3-06 Streaming & Event-Driven Pipelines 13 lessons 34-46 focused hours
Prerequisite	C3-05 Gate PASS; C2B streaming/lakehouse concepts may be recalled but Kafka/Spark streaming implementation is taught here.
Current lane	Apache Kafka 4.3.1 Spark Structured Streaming 4.2.0 Delta Lake 4.4.0 Debezium 3.6.2.Final
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny event example -> guided Kafka/streaming action -> expected result -> why it worked -> break/fix -> independent changed case -> validation -> replay/state/production reasoning -> evidence. A running stream is not mastery if duplicates, late data, offsets, checkpoints or sink state are wrong.

MeterStream scenario

Meter energy readings must feed near-real-time anomaly windows and a replayable lakehouse history.

Begin only after all 13 lessons and ShipmentStream Mini-Lab are complete. Lesson book, practice review and Mini-Lab review are closed during this assessment.

Independent requirements

- Inspect meter_events.jsonl; declare event_id, meter_id, event_time, ingest_time, schema version and business value field kwh.
- Design a Kafka topic with partition key, group, retention/replay policy and a single-broker learning boundary; collect real offset/group proof only if Kafka is actually run.
- Identify the duplicate event and define idempotent business-state behavior.
- Define one compatible schema evolution and one breaking change; state rollout/rejection behavior.
- Build an event-time window/watermark policy and classify the deliberately late event without overwriting source time.
- Complete a Structured Streaming/file-source or Kafka-source job with explicit schema and durable checkpoint. Literal Spark evidence is required only when the runtime is actually executed.
- Demonstrate same-checkpoint restart reasoning and a separately controlled replay/new-checkpoint path.
- Design/write a Bronze/Silver lakehouse sink contract with separate query checkpoints and deterministic idempotent commit/upsert behavior.
- Create an operator evidence packet: source count/unique IDs, duplicate/late handling, lag/backlog or query progress, state/checkpoint evidence, sink reconciliation and recovery runbook.
- Run one failure injection (duplicate, stale replay, schema incompatibility, checkpoint/sink issue) and prove recovery does not silently lose or duplicate accepted business state.

Competency Gate

Competency	Required PASS evidence
Architecture & event contract	Observed, reconciled, replay-safe, reproducible and explainable on this fresh case.
Kafka key/partition/group/replay reasoning	Observed, reconciled, replay-safe, reproducible and explainable on this fresh case.
Schema/CDC compatibility reasoning	Observed, reconciled, replay-safe, reproducible and explainable on this fresh case.
Duplicate/idempotency correctness	Observed, reconciled, replay-safe, reproducible and explainable on this fresh case.
Event-time/watermark/window correctness	Observed, reconciled, replay-safe, reproducible and explainable on this fresh case.
Checkpoint/restart/replay safety	Observed, reconciled, replay-safe, reproducible and explainable on this fresh case.
Lakehouse sink correctness	Observed, reconciled, replay-safe, reproducible and explainable on this fresh case.
Observability/runbook ownership	Observed, reconciled, replay-safe, reproducible and explainable on this fresh case.

PASS only if every required competency passes and zero unresolved critical failures remain. Runtime evidence that was not actually executed counts as missing evidence, not as a pass.

Evidence packet

Evidence	Your saved proof/path/note
Source/event contract + raw control totals	
Kafka topic/key/group/offset/replay evidence or explicit live-runtime pending	
Schema compatibility/CDC contract	
Duplicate/idempotency proof	
Event-time/window/watermark proof	
Checkpoint restart + fresh replay evidence	
Bronze/Silver sink reconciliation	
Lag/query-progress/state/sink monitoring	
Failure injection + recovery validation	
README/runbook + architecture defense	

Reviewer control values are intentionally excluded from this Gate brief. They appear only in the protected post-attempt review.